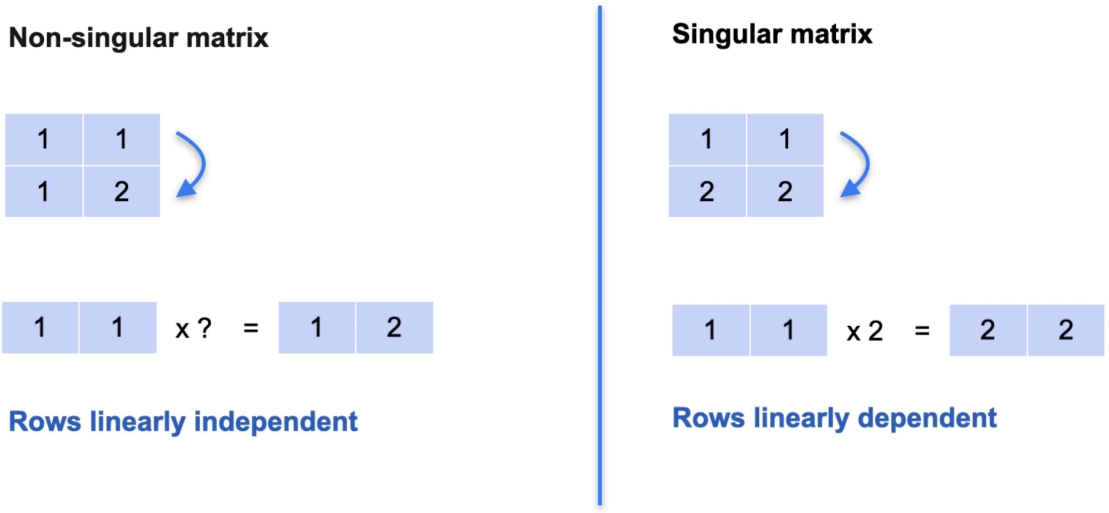
The determinant

It is a way to tell that the whether the matrix is singular or non-singular by calculating the determinant

The determinant = 0 --> the matrix is singular

The determinant is different from zero --> the matrix is non-singular

Dependency



a	b
c	d

Determinant $= ad - bc$

$$ak = c$$

$$bk = d$$

$$\frac{c}{a} = \frac{d}{b} = k$$

$$ad = bc$$

Matrix is singular if

a	b	* k =	c	d

Determinant

$$\boxed{ad - bc} = 0$$

Determinant of Square Matrix

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$|A| = a(ei - fh) - b(di - gf) + c(dh - eg)$$

In terms of Cofactor

$$|A| = a \times \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \times \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \times \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$